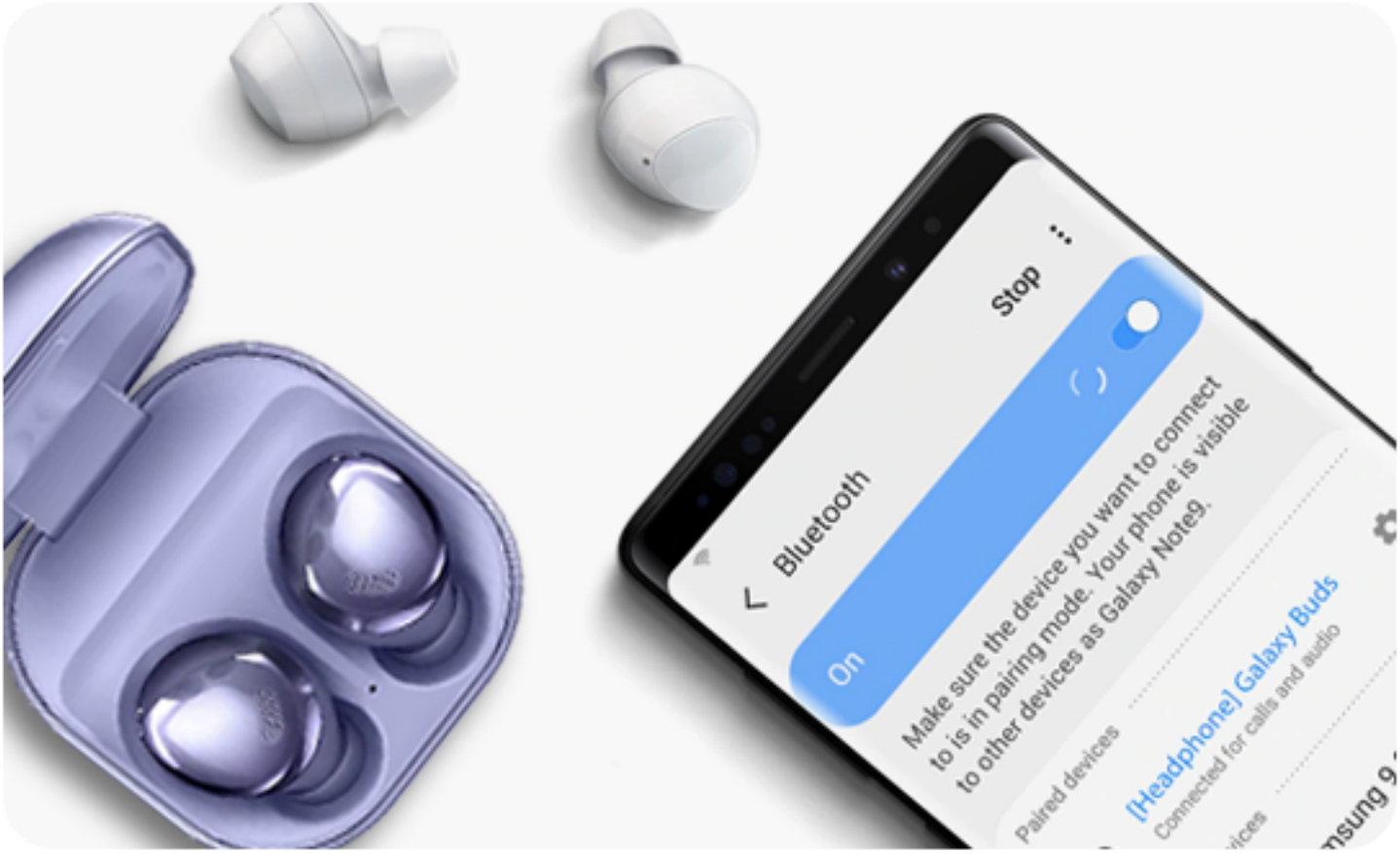
Types of Network

Types of Network in Computer Networks (Explained Clearly & In Detail)

Computer networks are classified mainly based on **geographical coverage area** and **purpose**.

1.PAN – Personal Area Network



A PAN is a small network used to connect personal devices within a short range (usually within 10 meters).

Range

Very small – around a room.

Example

- Mobile connected to Bluetooth earbuds
- Laptop connected to phone hotspot
- Smartwatch connected to phone

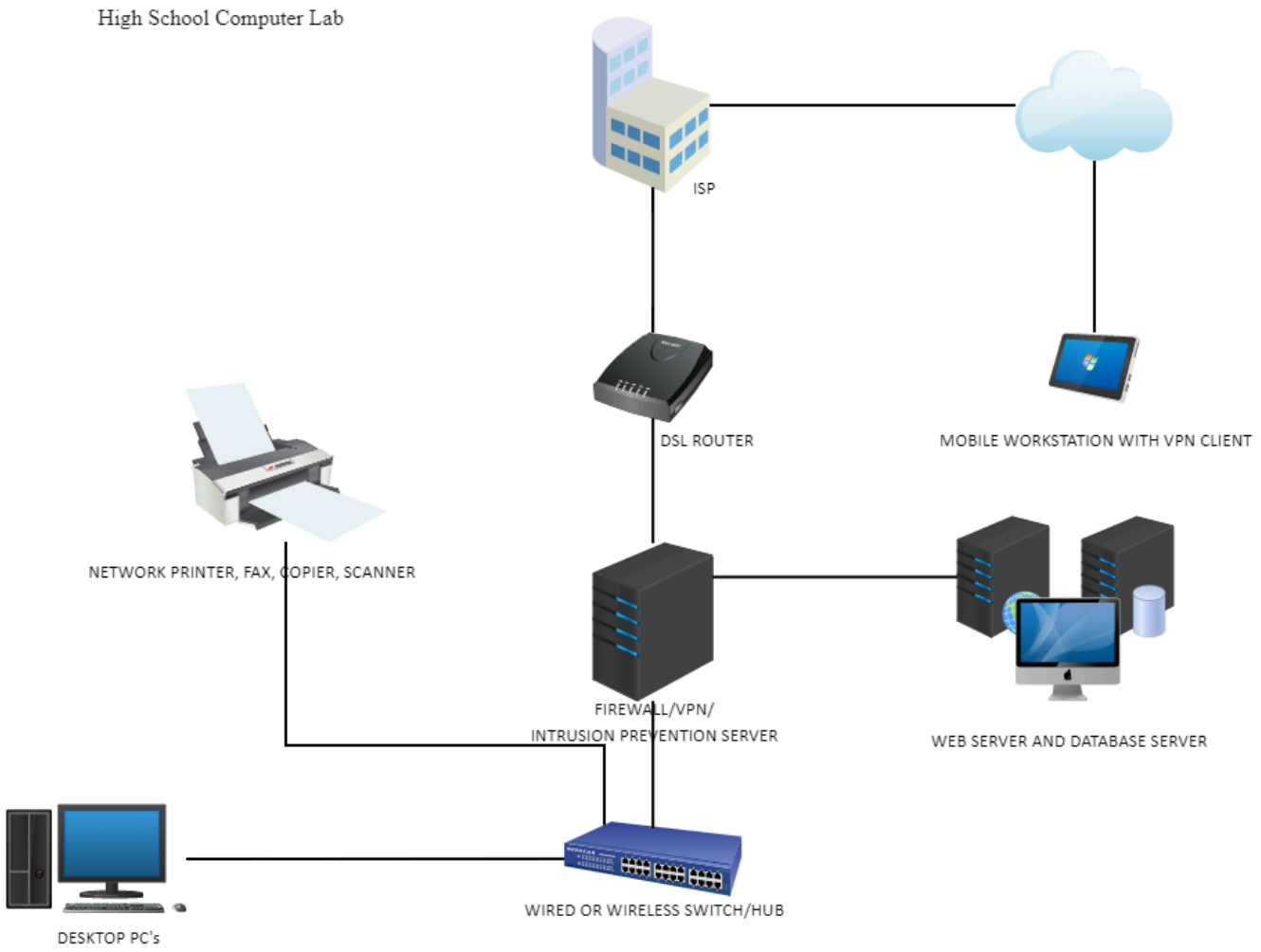
Technologies Used

- Bluetooth
- Infrared
- USB

Real-Life Use

When you share files from phone to laptop via Bluetooth.

2.LAN – Local Area Network



A LAN connects devices within a limited area like a home, school, office, or building.

Range

Few meters to a few kilometers.

Example

- School computer lab
- Office network
- Home WiFi network

Speed

High speed (100 Mbps to several Gbps)

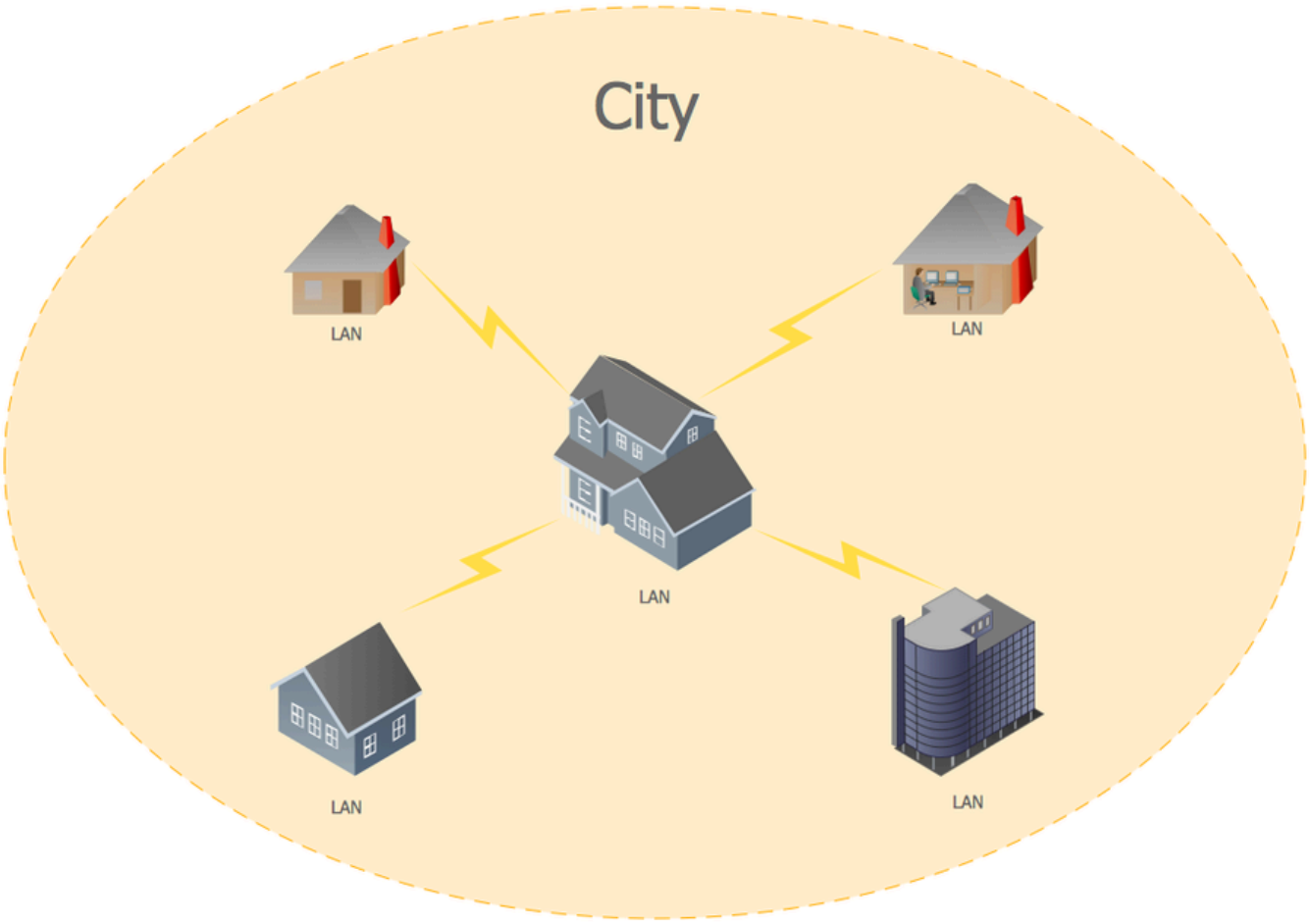
Devices Used

- Switch
- Router
- Ethernet cables

Advantage

Fast, secure, and low cost.

3.MAN – Metropolitan Area Network



A MAN covers a whole city or large campus.

Range

10–50 km

Example

- City-wide cable TV network
- ISP network across a city
- University campus network

Technology Used

- Fiber optics
- High-speed routers

4.WAN – Wide Area Network

A WAN connects networks over large geographical areas (countries or continents).

Range

Very large – thousands of kilometers.

Best Example

The Internet

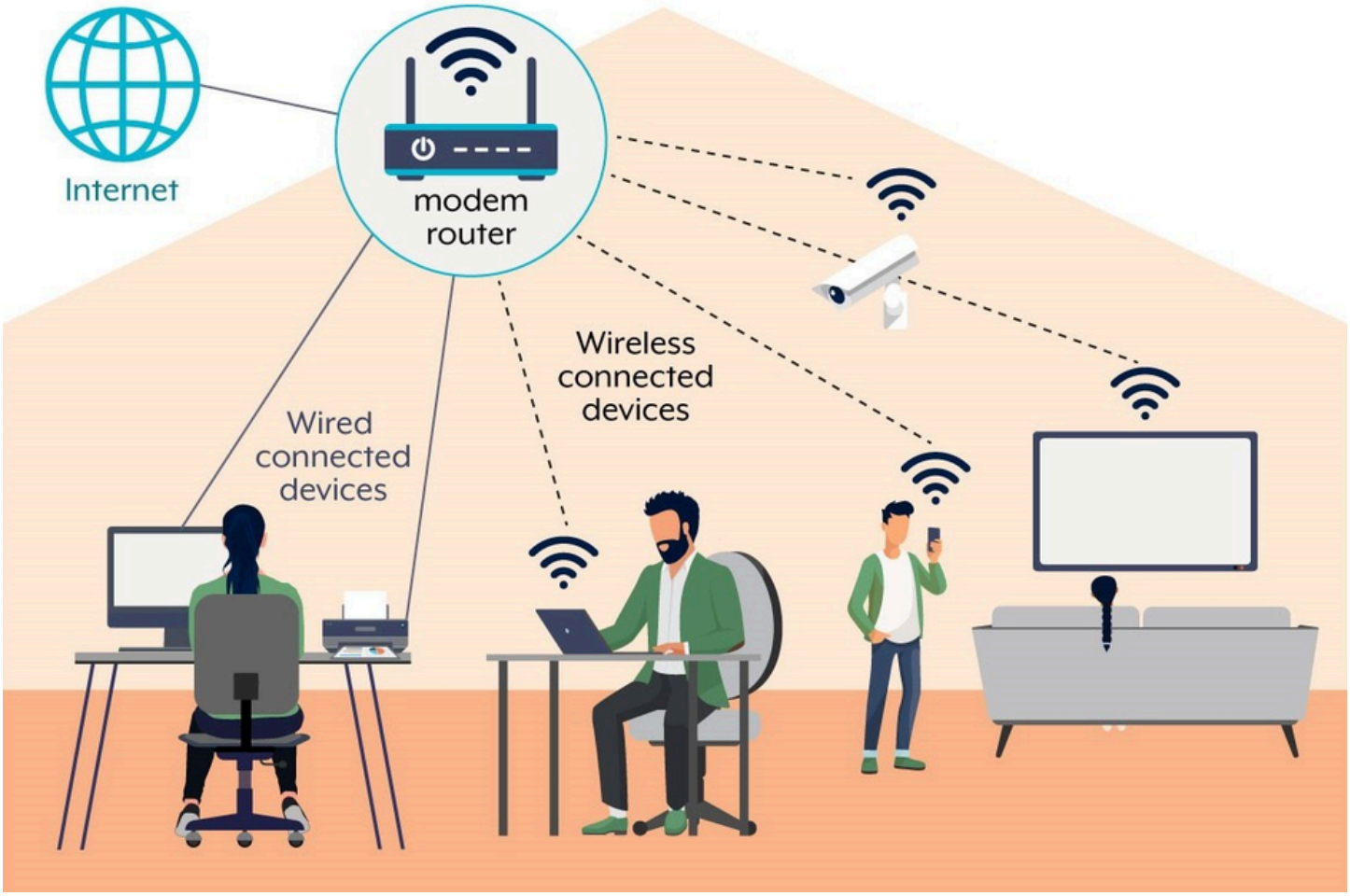
Example

- Bank network across multiple states
- Corporate network worldwide

Disadvantage

More expensive and complex.

5.WLAN – Wireless Local Area Network



A LAN that uses wireless technology (WiFi) instead of cables.

Example

- Home WiFi
- Coffee shop WiFi

Technology

- WiFi (IEEE 802.11)

6.SAN – Storage Area Network

A high-speed network that connects servers to storage devices.

Used In

- Data centers
- Cloud companies
- Large enterprises

Purpose

Fast data storage and backup.

Type	Full Form	Area Covered	Example
PAN	Personal Area Network	Very Small	Bluetooth
LAN	Local Area Network	Building	Office
MAN	Metropolitan Area Network	City	ISP
WAN	Wide Area Network	Country/World	Internet
WLAN	Wireless LAN	Building	WiFi
SAN	Storage Area Network	Data Center	Cloud Storage

Private IP Address in LAN & NAT

This is one of the most important concepts in Computer Networks.
Let's understand it step by step.

1.What is a Private IP Address?

A **Private IP address** is an IP address used **inside a local network (LAN)**.

It is **not visible on the Internet**.

It is used for communication between devices inside:

- Home network
- Office network
- School lab

Private IP Ranges

Class	Private IP Range
Class A	10.0.0.0 – 10.255.255.255
Class B	172.16.0.0 – 172.31.255.255
Class C	192.168.0.0 – 192.168.255.255

Most common home WiFi IP:

192.168.1.x

Example (Home Network)

You have:

- Laptop → 192.168.1.2
- Mobile → 192.168.1.3
- TV → 192.168.1.4

All these are **private IP addresses**.

They can talk to each other inside the house.

But these IPs **cannot be accessed directly from the Internet**.

2.Why Do We Need Private IP?

Because:

IPv4 addresses are limited
We cannot give public IP to every device
For security reasons

So we use:

- Private IP inside network
- Public IP for Internet

3.What is NAT? (Network Address Translation)

NAT is a technique used by the router.

It converts:
Private IP → Public IP
Public IP → Private IP

So devices inside LAN can access the Internet.

How NAT Works (Step-by-Step Example)

Situation:

You open YouTube on your laptop.

Laptop IP:

192.168.1.2

Router Public IP:

49.37.120.5

Step 1:

Laptop sends request to router:

From: 192.168.1.2
To: youtube.com

Step 2:

Router replaces private IP with public IP:

From: 49.37.120.5
To: youtube.com

This process is called **NAT**.

Step 3:

YouTube sends response to:

49.37.120.5

Router checks its NAT table and sends data back to:

192.168.1.2

You see YouTube video

4.NAT Table (Important Concept)

Router keeps a table like this:

Private IP	Public IP
192.168.1.2	49.37.120.5
192.168.1.3	49.37.120.5

Many private IPs use **one public IP**.

This is called:

PAT (Port Address Translation)
NAT Overloading

5.Types of NAT (Basic Idea)

1.Static NAT
One private IP → One public IP

2.Dynamic NAT
Private IP → Public IP from pool

3.PAT (Most common in home WiFi)
Many private IP → One public IP

6.Simple Real-Life Analogy

Think of:

Your House = LAN
Family Members = Devices
House Address = Public IP

When someone sends a letter, they send it to house address (public IP).

Inside house, father (router) gives letter to correct person (private IP).

That father's job = NAT.

7.Difference Between Private IP & Public IP

Feature	Private IP	Public IP
Visible on Internet	No	Yes
Used Inside LAN	Yes	No
Unique Globally	No	Yes
Example	192.168.1.2	49.37.120.5

Types of Network Topology

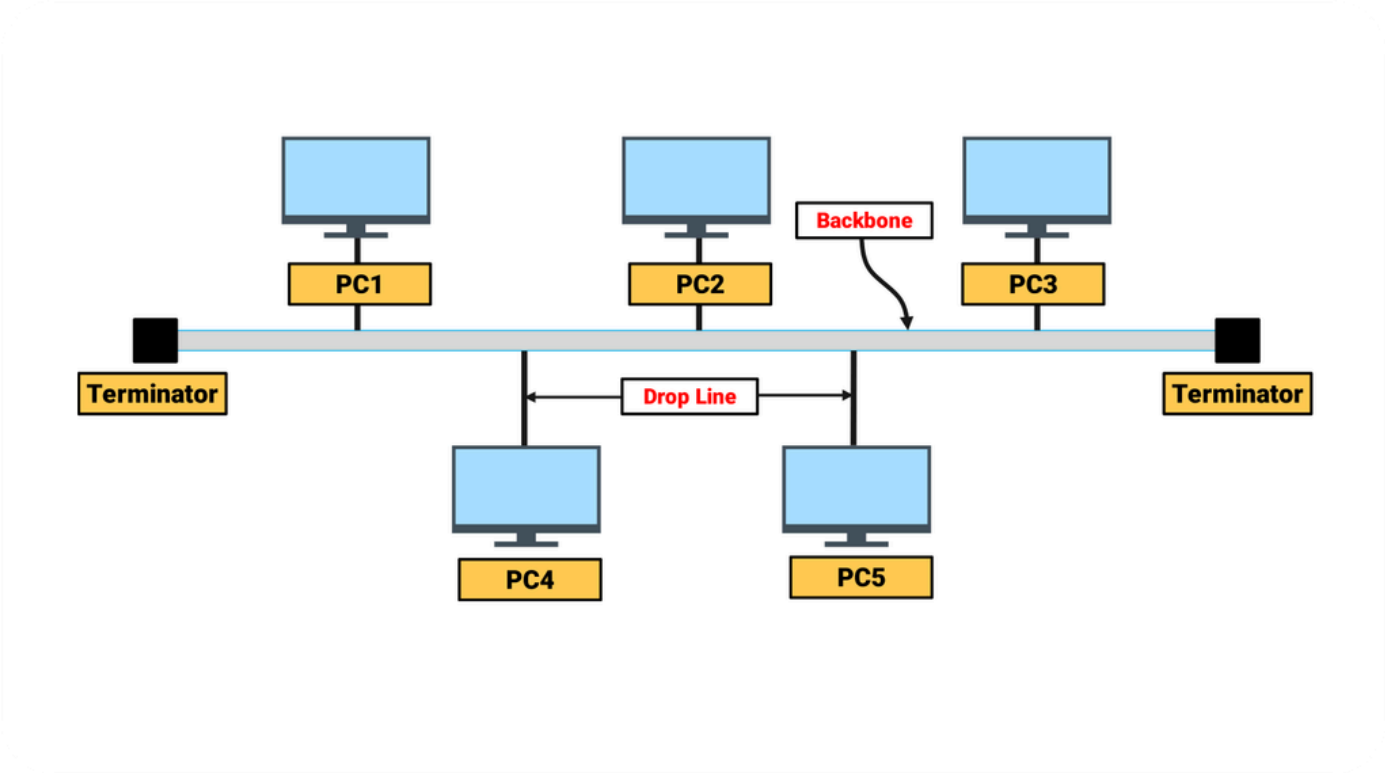
Network topology means **how computers are connected** to each other in a network.

Main types:

1. Bus
2. Star
3. Ring
4. Mesh
5. Tree
6. Hybrid

Let's understand one by one 👇

1.Bus Topology



What is Bus Topology?

All computers are connected to **one single main cable** (called backbone).

Data travels in one line.

Advantages

Simple and easy to install
Low cost (less cable required)
Good for small networks

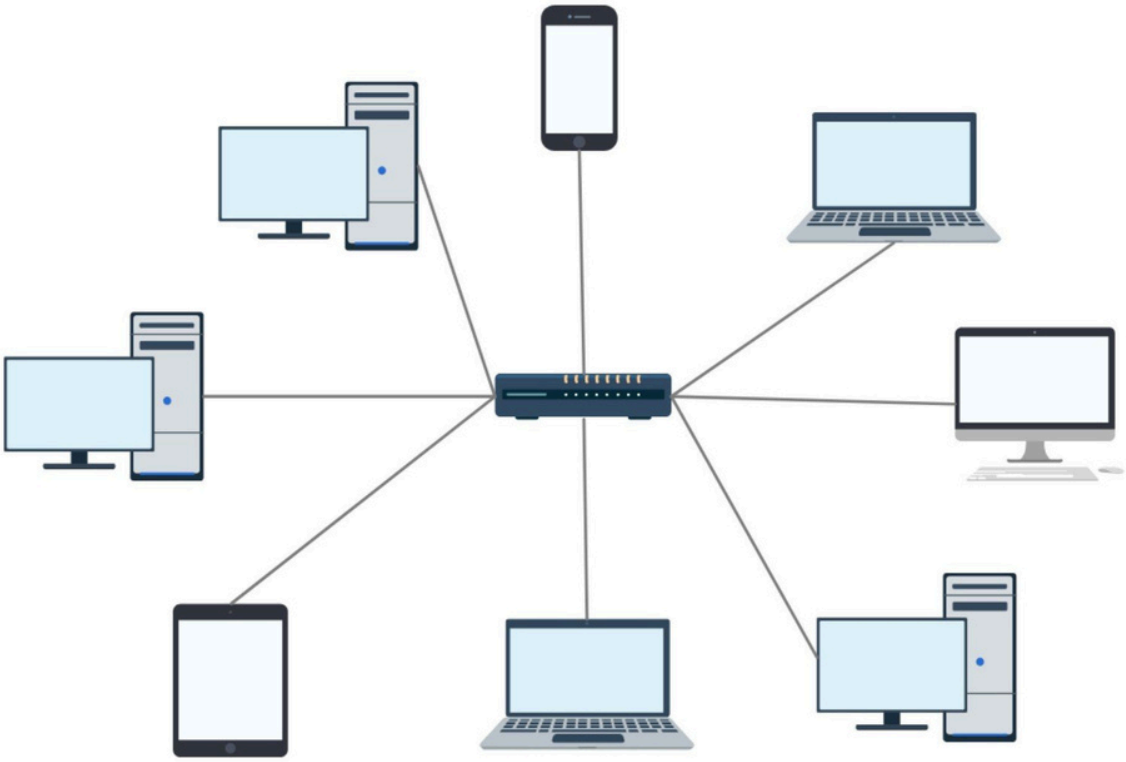
Disadvantages

If main cable fails → whole network stops
Performance decreases when devices increase
Hard to troubleshoot

Used In

Old small office networks.

2.Star Topology



What is Star Topology?

All devices are connected to a **central device** (Hub or Switch).

Data goes through the central device.

Advantages

Easy to install and manage
If one cable fails → only that device stops
Easy troubleshooting
High performance

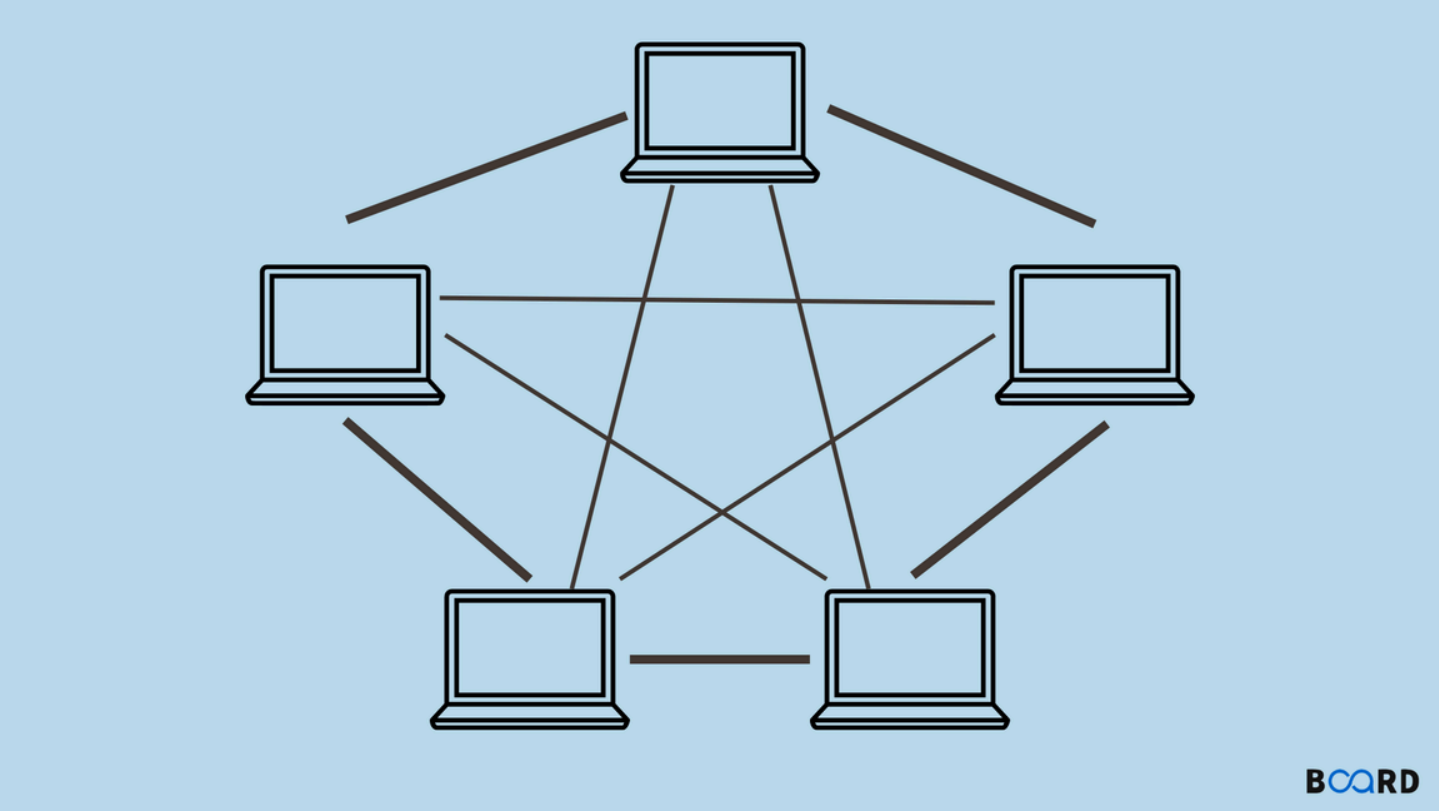
Disadvantages

If central hub fails → entire network stops
More cable required

Used In

Most modern LAN networks (Home & Office WiFi).

3.Ring Topology



What is Ring Topology?

Each computer connects to **two devices**, forming a circle.

Data travels in one direction.

Advantages

No data collision
Equal access for all devices

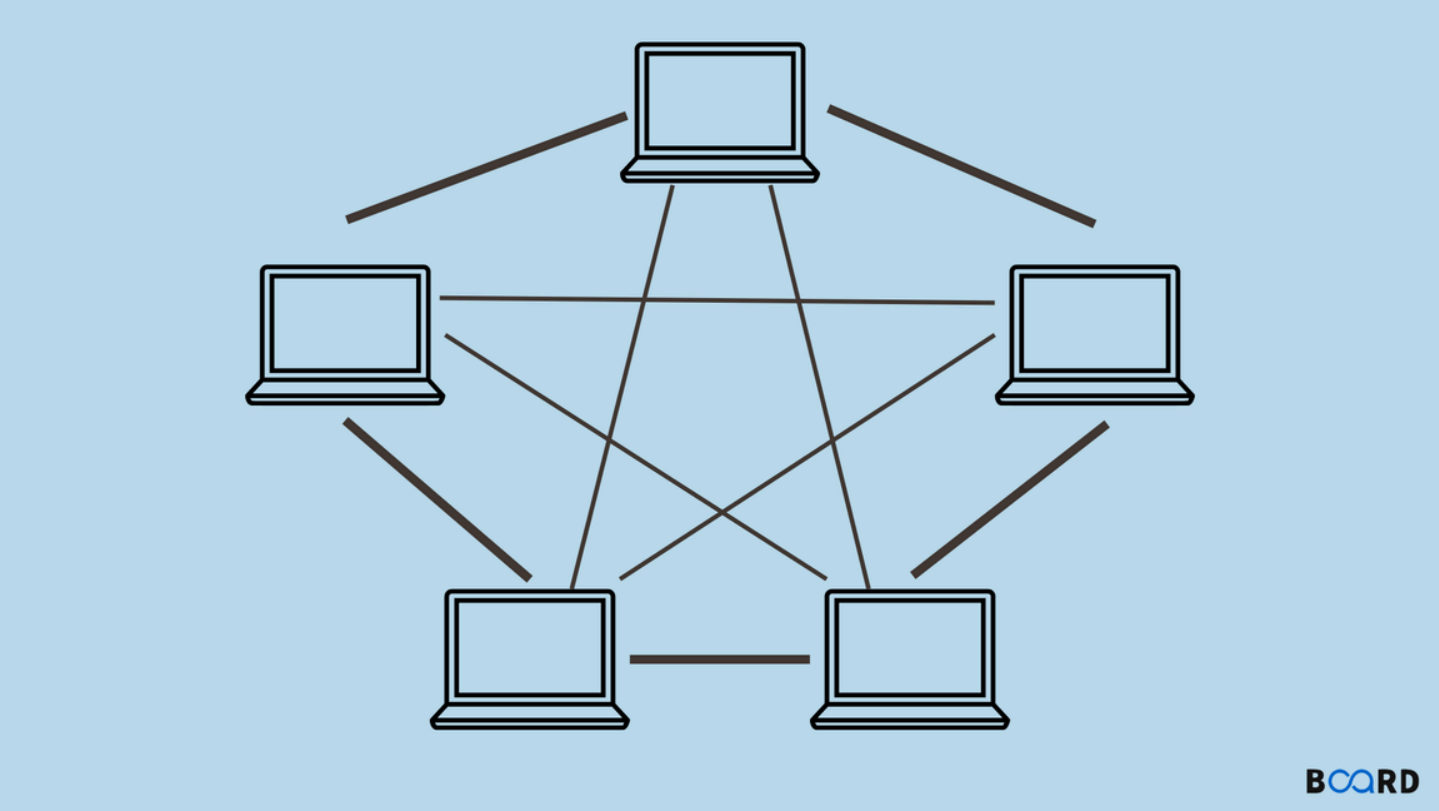
Disadvantages

If one device fails → network may stop
Hard to troubleshoot
Slower than star

Used In

Token Ring networks (older networks).

4.Mesh Topology



What is Mesh Topology?

Every device is connected to **every other device**.

There are multiple paths for data.

Advantages

Very reliable
If one link fails → data uses another path
High security

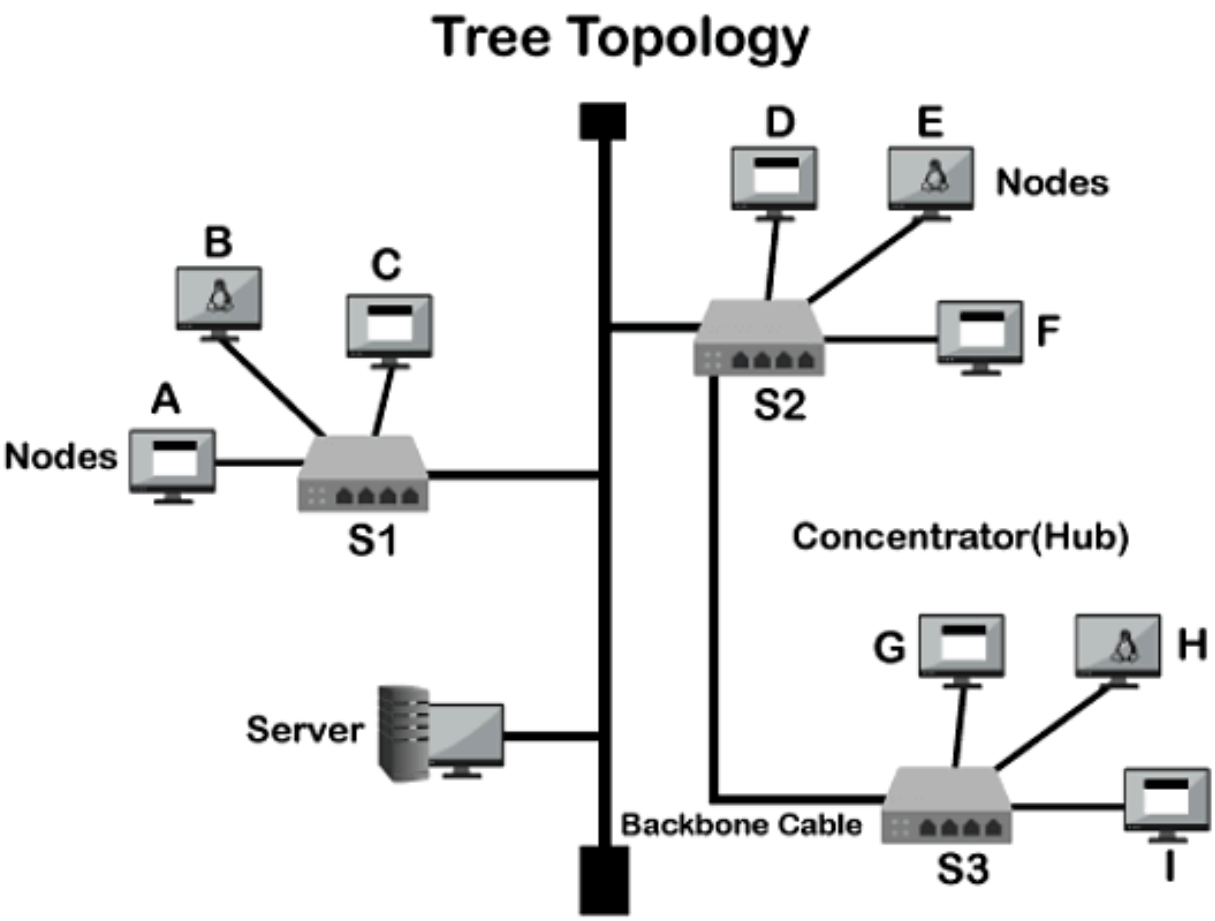
Disadvantages

Very expensive
Complex wiring
Hard to manage

Used In

Military networks, critical systems.

5.Tree Topology



What is Tree Topology?

Combination of **Star + Bus**.

It has a hierarchical structure (like a tree).

Advantages

Easy to expand
Good for large networks
Easy management

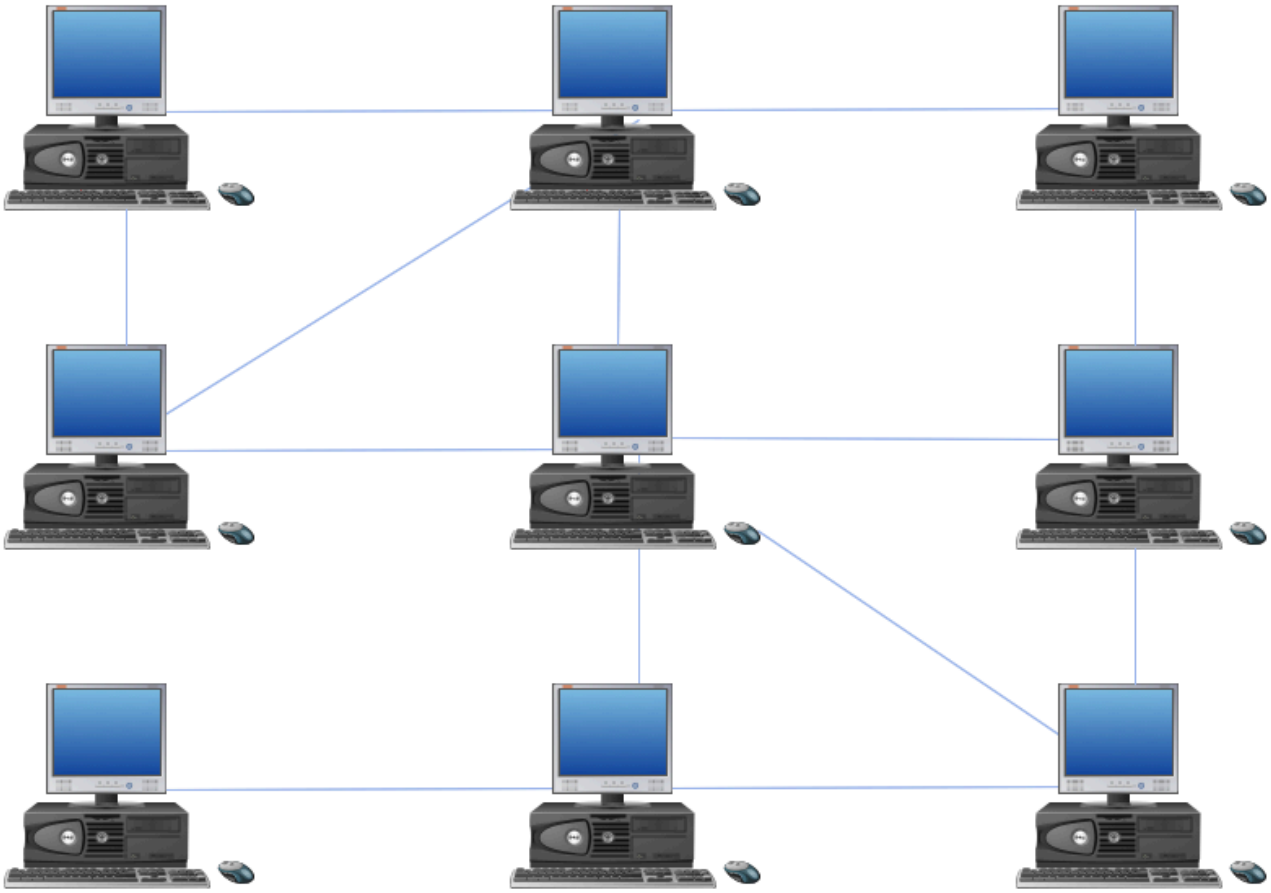
Disadvantages

If backbone fails → big problem
More cable required

Used In

Large organizations, universities.

6.Hybrid Topology



What is Hybrid Topology?

Combination of two or more topologies.

Example:
Star + Mesh
Star + Bus

Advantages

Flexible
Scalable
Reliable

Disadvantages

Expensive
Complex design

Topology	Cost	Reliability	Common Use
Bus	Low	Low	Small network
Star	Medium	High	LAN
Ring	Medium	Medium	Old networks
Mesh	High	Very High	Critical systems
Tree	Medium	High	Large org
Hybrid	High	Very High	Enterprise